

Phase 2 Requirement Analysis

FitBuddy AI Fitness Plan Generator

This phase converts FitBuddy's product direction into requirements that can guide design, development and review. The core acceptance target is a usable local flow that validates a profile, produces a structured week, saves it and supports a feedback-based revision.

Functional requirements

ID	Requirement	Acceptance check
FR 1	Collect profile details such as age, weight, goal, intensity, experience and equipment; allow movement notes where supported.	Valid input proceeds; blank or out-of-range required values receive understandable validation feedback.
FR 2	Generate a seven-day workout plan tailored to the supplied profile.	All seven days appear and use the requested equipment and experience level.
FR 3	Include warm-up, exercise guidance, recovery and safety notes.	Training days contain usable details and the week includes rest or active recovery.
FR 4	Provide a separate general nutrition or recovery tip.	A non-empty tip appears separately and is clearly not medical advice.
FR 5	Save a profile and generated plan in the local database.	The saved result can be retrieved through the application or supported API.
FR 6	Accept feedback and revise a saved plan.	The updated result is returned and the saved record reflects the latest version.
FR 7	Expose the main workflow through a browser interface and documented API.	A user can generate and review a plan in the web interface; API documentation is available locally.
FR 8	Provide an administrator view for saved records where enabled.	Unauthenticated access is rejected and valid local credentials permit access.

Quality requirements

- Safety: provide cautious general exercise guidance; do not diagnose, prescribe treatment or encourage training through pain.
- Usability: use clear labels, practical defaults, readable results and understandable validation or failure messages.
- Privacy: keep configuration and credentials private; avoid unnecessary sensitive information.
- Reliability: use the configured demo path where supported and show which generation mode produced a result.
- Maintainability: separate web handling, validation, AI generation and persistence.

Scope boundaries

The project is not a medical, rehabilitation or diet-treatment service. Clinical screening, wearable integrations, payments, public multi-user hosting and guaranteed fitness outcomes are outside the initial scope. Any later deployment needs a separate security, privacy and operational review.

Phase outcome

These requirements define review criteria for design and implementation. A result is incomplete if it omits days, ignores available equipment or cannot be retrieved after it is saved.